MIL-C-14264A
28 November 1978
SUPERSEDING
MIL-C-14264(CmlC)
28 February 1956

MILITARY SPECIFICATION

CALCIUM PHOSPHATE, TRIBASIC, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of tribasic calcium phosphate.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:01Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

ASTM D1193 - Reagent Water.
ASTM E11 - Wire-Cloth Sieves for Testing Purpose.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Physical and chemical characteristics. Tribasic calcium phosphate shall conform to table I when tested as specified therein.

TABLE I. Physical and chemical characteristics

Characteristic	Requirement		Test paragraph
	Minimum	Maximum	
Loss on ignition, percent by wt.	-	10.0	4.3.1
Assay, as P_2O_5, percent by wt.	39.8	41.2	4.3.2
Granulation, percent by wt:			4.3.3
Through No. 80 sieve	99.8	-	
Through No. 325 sieve	97.0	-	

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:01Z
Check the source to verify that this is the current version before use.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the tribasic calcium phosphate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 450 grams (g) shall be removed for each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1200	5
1201 to 7000	8
7001 to 20000	10
over 20000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one shipping container filled with tribasic calcium phosphate. Each unit shall be examined for the following defects using an AQL of 4.0 percent defective.

(a) Contents per shipping container not as specified
(b) Container incorrect or damaged
(c) Closure of container incorrect
(d) Marking incorrect, illegible, or incomplete.

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:01Z
Check the source to verify that this is the currentversion before use.

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.3. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.3 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.3.1 Loss on ignition. Ignite a porcelain crucible in a muffle furnace to constant weight. Weigh, to the nearest 0.1 milligram (mg), approximately 1.0 g of specimen into the tared crucible. Heat carefully over a Bunsen flame to prevent loss of specimen. Then, ignite in a muffle furnace at 800°C for 30 minutes. Cool to room temperature in a dessicator and weigh. Save the residue for assay, below. Calculate loss on ignition as follows:

$$\text{Percent loss on ignition} = \frac{100(B-T)}{A-T}$$

where: A = Weight, in grams, of specimen and crucible before ignition,
B = Weight, in grams, of specimen and crucible after ignition, and
T = Tare weight, in grams, of crucible.

4.3.2 Assay as P_2O_5.

4.3.2.1 Preparation of ammonium molybdate solution. Dissolve 65.0 g of finely powdered molybdic acid in a mixture of 140 milliliters (ml) of water and 145 ml of 28-percent ammonium hydroxide. Cool and add slowly to a well cooled mixture of 320 ml of 70-percent nitric acid and 400 ml of water. Allow to stand 48 hours and filter through an asbestos pad. Store the solution in a cool dark place. Use only the clear, supernatant liquid, filtering if necessary. (NOTE. This solution deteriorates on standing. If upon the addition of 2 ml of 1N sodium phosphate to 5 ml of the ammonium molybdate solution, an abundant yellow precipitate does not form at once or after slight warming, the ammonium molybdate solution should be discarded.)

4.3.2.2 Procedure. Dissolve residue from 4.3.2.1, above, in a mixture of 100 ml of water and 50 ml of 10-percent nitric acid. If necessary, filter the solution and wash the filter with several small portions of water. Transfer, with several rinsings, to a 250-ml volumetric flask. Pipet a 50-ml aliquot into a suitable size erlenmeyer flask. Add sufficient 10-percent ammonium hydroxide to produce a slight precipitate and then dissolve the precipitate by the addition

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:01Z
Check the source to verify that this is the current version before use.

of 1-ml of 10-percent nitric acid. Adjust the temperature to 50°C, add 75 ml of ammonium molybdate solution (4.3.2.1) and maintain the temperature at 50°C for 30 minutes, with occasional stirring. Allow precipitate to settle, and decant liquid. Add approximately 30 to 40 ml of water, mix well, and let settle. Decant liquid and repeat washing with a second 30 to 40 ml of water. Filter off liquid; wash precipitate with 1-percent potassium nitrate solution until washing is no longer acid to litmus paper. Transfer the precipitate and filter paper to the flask containing the washings. Add 40.0 ml of 1N sodium hydroxide and agitate until the precipitate dissolves. Titrate with 1N sulfuric acid to a colorless endpoint, using 3 drops of phenolphthalein solution as the indicator. Calculate the percent phosphorus pentoxide as follows:

$$\text{Percent } P_2O_5 = \frac{1.543(AB-CD)}{W}$$

where: A = Milliliters of sodium hydroxide solution,
B = Normality of sodium hydroxide solution,
C = Milliliters of sulfuric acid solution,
D = Normality of sulfuric acid solution, and
W = Weight, in grams, of residue after ignition (4.3.1).

4.3.3 Granulation. Tare, to the nearest 0.1 g, a No. 80 and a No. 325 sieve conforming to ASTM E11. Nest the No. 80 sieve on the No. 325 sieve on a receiving pan. Weigh, to the nearest 0.1 g, 100 g of specimen into the top sieve. Wash the specimen through the sieves by means of a stream of water from a faucet using a piece of rubber tubing, attached to the faucet, to direct the stream of water. (Care must be taken not to splash any of the specimen out of the sieve.) Continue the washing until no more specimen is washed through. Dry, to a constant weight, each sieve with residual specimen in a constant temperature oven at 105° ± 5°C. Cool to room temperature and reweigh. Calculate granulation as follows:

$$\text{Percent through No. 80 sieve} = \frac{[W - (X - A)]100}{W}$$ and

$$\text{Percent through No. 325 sieve} = \frac{[W - (X - A + y - B)]100}{W}$$

where: A = Tare weight, in grams, of No. 80 sieve,
B = Tare weight, in grams, of No. 325 sieve,
X = Weight, in grams, of No. 80 sieve and residual specimen,
y = Weight, in grams, of No. 325 sieve and residual specimen, and
W = Weight, in grams, of specimen.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:01Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packing, commercial. Tribasic calcium phosphate shall be commercially packed in accordance with MIL-STD-1188.

5.2 Marking. Marking shall be in accordance with MIL-STD-1188. Marking shall include lot number and date of manufacture.

6. NOTES

6.1 Intended use. Tribasic calcium phosphate, covered by this specification, is intended for use in colored smoke munitions.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - EA, AR
Air Force - 68
DLA - GS

Preparing activity:

Army - EA

Project No. 6810-0539

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:01Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-14264A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T03:01Z
Check the source to verify that this is the current version before use.

and dust frame and protected from light. (Refer to 7.3.2.)

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commander
US Army Armament Research
and Development Command
Attn: DRDAR-TSC-S
Aberdeen Proving Ground, MD 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-18T03:01Z
Check the source to verify that this is the current version before use.